

Actualities: Seamless Live Performance with the Physical and Virtual Audiences in Multiverse

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ABSTRACT

During the pandemic, the popularity of VR concerts increased, providing the audience with a novel experience of watching performances where performers and online attendees typically exist in different spaces and times. However, as the pandemic recovered gradually, people returned to live performances physically due to the diminished sense of co-presence experienced in VR concerts. We present Actualities, a seamless live performance consisting of onsite and online form with a multiverse interaction to create an engaging experience for all participants. By utilizing various sensors, we capture signals from the performance venue and digitalize all onsite elements into virtual scenes. The visual content is projected onto screens for the onsite audience, while being simultaneously broadcast via live-streaming to the online audience.

CCS CONCEPTS

• Human-centered computing → Interaction design.

KEYWORDS

Multiverse Interaction, XR, Live Performance, Audio-Visual

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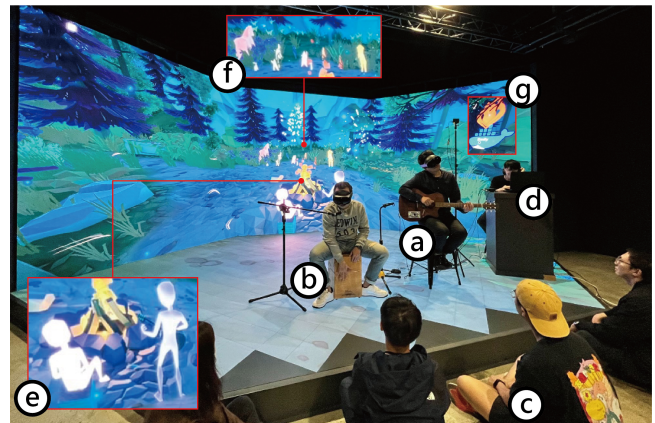


Figure 1: The configuration for a seamless live performance: (a)Guitarist. (b)Cajon drummer. (c)Onsite audiences. (d)VJ. (e)Performers' avatars. (f)Onsite audiences' avatars. (g)Emoji feedback from Online audiences.

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1 INTRODUCTION

People tend to participate in a live concert and share an unique experience with each other [Brown and Knox 2017], indicating that audience engagement is an important element of a live performance. Recent technological advances offer audiences new ways to attend live concerts, which is with virtually attending gigs and live-streaming, individuals can watch concerts while physically alone, but virtually together [Charron 2017]. VR concerts are becoming more common nowadays, and VR concert platforms seek to recreate the live concert experience in the virtual world.

In 2019, the band Miro Shot held a XR concert and provided audiences with multi-sensory experiences by controlling parameters such as the wind, temperature and touch [Amirsadeghi 2020]. In 2020, Travis Scott partnered with Fortnite, one of the biggest metaverse platforms, for a unique in-game concert. [Horie et al. 2018] present a VR concert experience system in which visual effects appear depending on a user's brain waves and heartbeats. VR concerts have distinct experiences from in-person concerts, such as non-physical space, immersive 360-degree views, and animated environments that can be customized to create unique and dynamic visuals. However, [Yakura and Goto 2020] indicate that VR concerts lack co-presence, and examine computational approaches to enhance the sense of co-presence through virtual audience avatars.

In this work, we present Actualities, a XR concert system that provides physical and virtual audiences with a seamless live performance experience in the multiverse. Whether the audience attends physically or virtually, they can sense and interact with each other.

2 DESIGN AND IMPLEMENTATION

Several sensors are used to capture the signals during the live performance. For the guitar, we use LiCAP, an arrayed optical pickup, to digitize the vibration signals from the strings. When the guitarist strums the guitar, flowers and butterflies will be generated in real time. As for the cajon drum, we use capacitive sensors to digitize the striking signals. The particle explosion shows up simultaneously as the drummer strikes the cajon. The music visualization is generated at the same time as the two musicians play the instruments (Figure 1e), cocreating an audio-visual content. On the other hand, millimeter wave radar sensors are used to detect the movement of the onsite audiences, and digitalize them into virtual avatars (Figure 1f). All the real-time visuals are completed in a 3D scene built with Unity and projected on the semi-immersive screen during the concert. The onsite performance will be broadcast simultaneously to the online audiences through 360 livestream system, and they can experience from three viewpoints with their mobile devices (Figure 2b). Online audiences can also send emoji feedback in the livestreaming platform with mobile devices (Figure 2a), creating visual effects showing on the onsite screen (Figure 1g). The performers can see the onsite audiences with Quest Pro via video see-through, and there is a window shown on their headset which enables them to watch the virtual environment and the online audiences' emoji feedback (Figure 2c).

3 SEAMLESS LIVE PERFORMANCE

We organized a XR live performance consisting of onsite and online concerts synchronously, which allows both physical and virtual audiences to coexist and interact within a virtual environment. We created a virtual setting, presenting a forest concert where musicians performed three melodies, each conveying a different style: happiness, romance, and sadness. The scenario transitioned from daytime to dusk and finally to night, while the music visualization dynamically adjusted its color tones based on the music and background. During the performances, real-time music visualization was displayed on a semi-immersive screen, while physical attendees were transformed into digitalized representations of wild



Figure 2: Viewpoints from different users: (a)Onsite audiences' viewpoint. (b)Online audiences' viewpoint. (c)Viewpoint from performers wearing Quest Pro.

animals, joining the forest concert as part of the performance. Online audiences expressed their feedback through emoji, which was incorporated into the virtual environment. Every participants became an integral part of the performance, assuming various forms and engaging with each other within the multiverse.

4 DISCUSSION AND FUTURE WORK

This work explores the possibilities of seamless interaction between physical and virtual audiences in multiverse, and we provide a XR live performance to achieve our goal. In this work, we use several sensors to capture the signals from the instruments and the onsite audiences, and transform the information into digital versions in the virtual environment. Physical audiences and virtual audiences experience the performance in different actualities, however, they can still sense the co-presence and interact with each other. In the future, we will carry out user studies to investigate the presence, immersion, and audience participation in this XR performance.

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